

URŠA URŠIČ (SHE/HER)

Graduate student in experimental cellular biophysics

Max Planck Institute of Molecular Cell Biology and Genetics, Dresden, Germany

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in ursaursic

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My scientific mission is to expand human knowledge about the basic principles of life using approaches at the interphase between physics and biology. I love to collaborate with people from various scientific backgrounds on novel projects and creatively solve problems along the way. My doctoral project includes understanding how material properties of the cytoplasm affect cell division mechanisms in large embryonic cells.

Experience

📅 Jun – August 2024

Physiology course

Marine Biological Laboratory

📍 Woods Hole, MA, USA

Rotations: 1. Extremotolerance of *A. Pullulans* (mentor: Amy Gladfelter), 2. Phagocytosis with surface modifications of macrophages and targets. (mentor: Daniel Fletcher) 3. Biomineralization of ossicles in sea cucumbers. (mentor: Manu Prakash)

📅 Jun – August 2022

Internship

Flatiron Institute, Simons Foundation

📍 New York, New York, USA

Conducted computational research in microtubule self-organization using the innovative simulation tool, aLENS, developed within the group.

Mentors: Adam Lamson & Michael Shelley

📅 Nov 2021

Predoc course

Max Planck Institute of Molecular Cell Biology and Genetics

📍 Dresden, Germany

A 4-week group research project on the shininess of Zebrafish. Conducted experimental as well as computational research on guanine crystal formation in iridophores in Zebrafish skin.

Mentors: Rita Mateus, Eric Geertsma, Carl Modes

📅 Oct 2020 – Sep 2021

Master's thesis

F5 – Institute 'Jozef Stefan', Faculty for Mathematics and Physics – UL

📍 Ljubljana, Slovenia

Performed experiments on lung epithelial cells and observed their mechanical response to osmotic shock. Observed that exposure to TiO_2 nanoparticles heightened the response to osmotic shock, indicative of cytoskeleton weakening.

Mentors: Iztok Urbančič, Miha Ravnik

📅 Jun – Jul 2019

Internship

Protein, Structure and Function Group, University of Queensland

📍 Brisbane, Australia

Characterization of innate immune system protein aggregation (TLR4 and MAL) using fluorescence confocal spectroscopy (FCS) techniques and cell-free protein expression techniques. Improved and automated analysis of the FCS data output.

Mentor: Boštjan Kobe

📅 Oct 2018 – Apr 2019

Internship

ESMP – University of Luxembourg

📍 Luxembourg, Luxembourg

Laser-induced motion of topological defects on nematic liquid crystal shells. Production of liquid crystal shells with microfluidics and using a custom-made double confocal microscope imaging.

Mentor: Jan Lagerwall

📅 Feb 2017 – Sep 2018

Student project

K5, F5 – Institute 'Jozef Stefan'

📍 Ljubljana, Slovenia

Characterization of the hydrodynamics of drying droplets for inkjet printing of flexible electronic circuits.

Production of metal oxide thin films and characterization of their temperature-dependent dielectric constant.

Mentor: Aleksander Matavž

Education

📅 Oct 2021 – in progress

Ph.D. in Biophysics

Max Planck Institute of Molecular Cell Biology and Genetics

📍 Dresden, Germany

📅 Oct 2019 – Sep 2021

M.Sc. in Biophysics

University of Ljubljana

📍 Ljubljana, Slovenia
grade 9.5/10

📅 Oct 2015 – Apr 2019

B.Sc. in Physics

University of Ljubljana

📍 Ljubljana, Slovenia
grade 8.6/10

Teaching & Mentoring

📅 Winter semester 2024

Advanced Practical: Biomolecular motors

3rd year Physics students

📍 TU Dresden, B-Cube

📅 June 2023

Biophysical modelling workshop

📍 Flatiron institute, New York
Teaching assistant for aLENS (1/3 of the full course)

📅 Winter semester 2022

Advanced Practical: Biomolecular motors

3rd year Physics students

📍 TU Dresden, B-Cube

📅 Winter semester 2022

Molecular Machines Practical

Masters course in Bioengineering

📍 TU Dresden, MPI-CBG

📅 2016-2018

Private saxophone lessons

📍 Slovenija

Publications

- Kickuth A., Uršič, U., Staddon, M., Brugues J., 2026, *Nature*
- Matavž, A., Uršič, U., Močivnik, J., Richter, D., Humar, M., Čopar, S., Malič, B. and Bobnar, V., 2022. *Journal of Colloid and Interface Science*, 608, pp.1718-1727.

Talks & Presentations

- 2024 ASCB poster
- 2024 ASCB talk
- 2024 POL seminar
- 2023 Internal MPI-CBG seminar
- 2023 Invited talk at Slovene Biophysical Society
- 2023 Chalk talk at Science on Board at MPI-CBG
- 2022 Poster presentation at Flatiron Institute summer internship

Awards & Fellowships

- ★ 2024 Marine Biological laboratory postcourse research fellowship
- ★ 2024 Society for General Physiology Award for Scholars at MBL
- ★ 2024 CoolLED image competition
- ★ 2022 Simons Foundation summer intern fellowship
- ★ 2021 Dean's Award for top 5% of students at the Faculty of Mathematics and Physics, University of Ljubljana
- ★ 2011-2021 National 'Zois' scholarship for gifted students
- ★ 2021 Silver national award in competition in logics
- ★ 2020 BioCamp 2020 Fellow (Lek, Novartis)
- ★ 2019 Silver National Award in logics competition
- ★ 2019 ASEF Junior Fellowship

Languages

Slovenian
English
Russian
German



Side projects

- Member of VTIS (2023-recent),
- Volunteer at Science goes to school (2023-2024)
- Volunteer at Moja Mavrica - safe space for the LGBT community with a focus on mental health (2022-2023),
- Volunteer at DIH - Peer support for LGBT youth (2021-2023),
- An active member of ASEF - buddy program, tutoring group, activities (2019-2022),
- Clean water round table, ASEF (2021)

Hobbies

Music (saxophone, piano)

Dancing

Hiking

Running

Drag performances

Volleyball